Academic Elective Bidding & Allocation System (Problem Statement #05)

Deliverable 3: Use-Case Flow Specification (1-Page Core Specification)

Use Case ID	**UC-01**	**Use Case Name**	**Submit & Manage Elective Bids**
Primary Actor	Student	**Secondary Actor**	Academic Database / Prerequisite Service
Priority	**High**	**Trigger**	Student accesses elective bidding module during active window
Description	Enables a student to allocate 100 bidding credits across ranked elective choices, validate prerequisites, and lock their bidding submission for algorithmic allocation.		
Preconditions	1. Student is authenticated with active enrollment status. 2. Elective bidding window is currently active/open. 3. Student has exactly 100 unspent bidding points available for the semester.		
Postconditions	1. Student's elective bids and credit allocations are persistently stored in the database. 2. 100 bidding points are marked as allocated. 3. An immutable audit log record and confirmation receipt are generated.		

Main Success Scenario (Step-by-Step Flow):

1. **Student** logs into the portal and navigates to the 'Elective Course Bidding' section.
2. **System** retrieves and displays the catalog of offered elective courses, descriptions, available seats, faculty, and designated timetable slot timings.
3. **Student** selects desired elective courses and distributes exactly 100 bidding points among them based on personal priority.
4. **Student** submits the bidding form by clicking 'Submit Bids'.
5. **System** invokes «include» **UC-02: Validate Course Prerequisites** to verify student's transcript eligibility.
6. **System** validates that the sum of allocated bidding points equals exactly 100.
7. **System** verifies that no two selected electives share conflicting mandatory core slot schedules.
8. **System** records the bid distribution in the database and generates an immutable audit timestamp.
9. **System** displays a confirmation dialog: *'Your elective bids have been successfully recorded with 100 points distributed.'* and generates a downloadable confirmation summary.
10. **Use Case ends successfully.**

Alternate Flows & Exception Handling:

4a. Bidding Points Total Mismatch (≠ 100 Credits)	**5a. Unmet Course Prerequisites**
• **4a1.** System detects sum of allocated credits ≠ 100 (e.g., total = 90 or 110). • **4a2.** System blocks submission with error: *'Total allocated points must equal exactly 100 credits.'* • **4a3.** System highlights point balance counter. • **4a4.** Student adjusts values to 100 and resubmits (re-enters Step 4).	• **5a1.** System detects student has not passed prerequisite Course P for Elective E. • **5a2.** System rejects bid for Elective E: *'Ineligible: Prerequisite [P] not met.'* • **5a3.** System prompts student to remove or replace Elective E. • **5a4.** Student updates selection, readjusts points, and resubmits.
7a. Timetable Slot Collision («extend» UC-03)	**8a. Bidding Window Closes During Submission**
• **7a1.** System detects two selected electives share the exact same timetable slot. • **7a2.** System triggers **UC-03: Resolve Timetable Conflict**, warning student that only one can be allotted. • **7a3.** Student designates primary vs backup preference and resumes at Step 8.	• **8a1.** System checks timestamp; bidding deadline has passed before DB commit. • **8a2.** System aborts transaction and displays: *'Bidding window has closed.'* • **8a3.** Use case terminates without state change.